Can Edge Addition Be Safe and Effective? Adjacency-Centered Augmentation via Langevin and SDE Diffusion for Self-Supervised Graph Anomaly Detection

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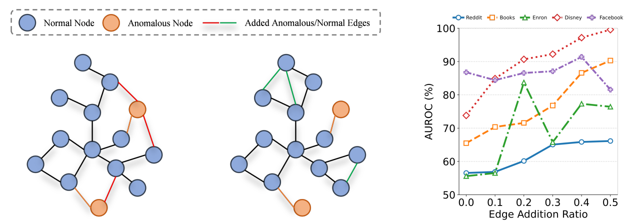
Abstract

Edge addition is commonly considered risky in Graph Anomaly Detection (GAD), as random edge addition may induce anomaly–normal connectivity. Consequently, most existing augmentation strategies focus on feature perturbation, edge removal, or subgraph sampling, leaving edge addition largely unexplored. In contrast, we empirically find that moderate and targeted structural completion among normal nodes consistently improves GAD performance, revealing guided edge addition as an overlooked yet effective augmentation dimension. Motivated by this observation, we introduce two adjacency-centered edge generation strategies with complementary mechanisms. One performs a training-free structural completion scheme via spectrum-aware Langevin dynamics, enriching graph connectivity while preserving node features. The other models the joint evolution of node features and graph structure through a stochastic differential equation–based diffusion process, producing structurally coherent and anomaly-aware complementary graphs. Extensive experiments on 12 benchmark datasets with 7 state-of-the-art GAD models demonstrate consistent and substantial improvements in both AUROC and AUPRC. Code and appendix are available online.[1]



(a) Unhealthy Connections (b) Healthy Connections (c) Effect of Connection Ratio on BOURNE

Figure 1: A toy illustration of healthy vs. unhealthy connections, and the performance gains of BOURNE on five real-world anomaly datasets with varying normal–normal connection ratios.

1 Introduction

Graph Anomaly Detection (GAD) aims to identify nodes in static graphs that deviate significantly from the majority in real-world information networks. Although such anomalies are typically rare, their detection is crucial due to the potential economic and social consequences. Representative examples include fraudulent users on online shopping platforms [Yu *et al.*, 2024; Zhang *et al.*, 2022], fake news on social media [Feng *et al.*, 2025; Sun *et al.*, 2025], spam on review-sharing platforms [Deng *et al.*, 2022], and intruders in communication networks [Westphal *et al.*, 2025]. With the increasing availability of graph-structured data and the rapid advancement of deep graph neural networks, GAD has attracted substantial research interest across multiple disciplines [Ma *et al.*, 2025; Ma *et al.*, 2021], and has become an important problem in both theoretical research and practical applications.

Existing GAD methods are fundamentally challenged by the severe class imbalance between normal and anomalous nodes, which biases the learning process. Prior work addresses this issue through model design and data augmentation, with prevailing strategies relying on node feature masking, random edge dropping, and subgraph sampling, and recent extensions incorporating diffusion-based graph generators to enrich anomaly patterns. Despite these advances, edge addition or structural completion is rarely explored in GAD augmentation. The primary concern is that introducing new edges may contaminate anomaly-sensitive representations by creating spurious connections involving anomalous nodes, thereby degrading detection performance, as illustrated in Fig. 1(a).

However, as shown in Fig. 1(b) and (c), we empirically observe that carefully increasing edges among normal nodes consistently improves the performance of existing GAD models, with gains scaling approximately proportionally to the number of added edges. This suggests that guided edge addition preserving intra-community connectivity enhances feature aggregation among normal nodes while suppressing the influence of anomalous nodes. These results reveal an underexplored yet promising direction for graph augmentation and naturally raise the question of how to design a principled, label-free strategy for guided edge addition that reliably improves GAD.

To assess the feasibility of guided edge addition in real-world GAD scenarios, we revisit graph denoising and diffusion-based generative models from a spectral energy

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[1]https://github.com/GaoHaokai222/LangGen-JointSDE

perspective and examine their applicability to anomaly detection. Leveraging the relationship between graph signal frequencies and anomalous connectivity patterns, we propose two adjacency-matrix-centered diffusion-based augmentation methods, LangGen and JointSDE. LangGen is a feature-independent, training-free approach based on Langevin dynamics, which integrates the Boltzmann distribution with the Rayleigh quotient to increase edge density while preserving low-frequency spectral energy, thereby strengthening intra-community connectivity without model training. JointSDE, in contrast, formulates a joint feature–structure diffusion process via stochastic differential equations, where spectral graph principles guide the synchronized evolution of node features and the adjacency matrix. This joint diffusion gradually densifies graph structure while regularizing feature distributions, enabling the reconstruction of edge-augmented graphs whose structural and feature representations are jointly evolved and mutually consistent. Extensive experiments on multiple benchmarks with state-of-the-art GAD models demonstrate that both methods generate high-quality edge-augmented graphs with efficient computational and memory overhead, leading to consistent improvements in AUROC and AUPRC.

The contributions of this work are summarized as follows:

- To the best of our knowledge, this work is the first to explore edge addition as a graph augmentation strategy for graph anomaly detection, showing that carefully increasing edges among normal nodes can improve the performance of diverse GAD models.
- Two adjacency-matrix-centered, spectral-guided diffusion methods, LangGen and JointSDE, are proposed to generate high-quality edge-augmented graphs without label information.
- Extensive evaluations on seven state-of-the-art GAD models across twelve benchmark datasets validate the effectiveness, efficiency, and broad applicability of the proposed augmentation strategies.

2 Related Work

2.1 Graph Anomaly Detection

Graph anomaly detection aims to identify rare patterns that deviate from dominant graph structures or attributes [Ma *et al.*, 2021; Pazho *et al.*, 2023; Wang *et al.*, 2025; Jin *et al.*, 2024]. Early methods such as Radar [Li *et al.*, 2017] and ANOMALOUS [Peng *et al.*, 2018] detect anomalies via matrix factorization and residual analysis. Deep learning-based approaches, including DOMINANT [Ding *et al.*, 2019], AnomalyDAE [Fan *et al.*, 2020], and ComGA [Luo *et al.*, 2022], detect anomalies via reconstruction errors in graph autoencoders, under the assumption that anomalous nodes are harder to faithfully reconstruct from learned normal patterns. Contrastive learning methods, such as CoLA [Liu *et al.*, 2021], SL-GAD [Zheng *et al.*, 2021], HUGE [Pan *et al.*, 2025], and SAMCL [Hu *et al.*, 2025], enhance detection by contrasting node- and subgraph-level representations. Recently, inspired by the observation that anomalies manifest distinctive spectral energy patterns [Tang *et al.*, 2022], spectral-based GAD has emerged as an active research direction [Gao *et al.*, 2023; Dong *et al.*, 2024; Xu *et al.*, 2024; Zheng *et al.*, 2025]. Representative methods such as UniGAD [Lin *et al.*, 2024] unify node-, edge-, and subgraph-level detection through subgraph modeling with Maximum Rayleigh Quotient (MRQ)–based sampling, while SALAD [Ma *et al.*, 2025] and ADA-GAD [He *et al.*, 2024] mitigate anomaly interference in normal structure learning through loss reweighting or Rayleigh energy–based denoising. Despite these advances, existing GAD methods predominantly rely on limited and fixed augmentation strategies, such as node masking, feature perturbation, edge dropping, and subgraph sampling, while the potential of edge addition or structural completion remains unexplored.

2.2 Diffusion Model

Diffusion models [Ho *et al.*, 2020; Song *et al.*, 2021a; Song *et al.*, 2021b] have recently emerged as powerful generative frameworks, achieving strong performance across image synthesis [Huang *et al.*, 2025], time-series modeling [Yang *et al.*, 2024], and other domains [Cao *et al.*, 2024]. This success has motivated their extension to graph-structured data. DiGress [Vignac *et al.*, 2023] introduces a discrete diffusion process that perturbs graph structures by stochastically modifying edges and node attributes, and learns an inverse transformation to recover clean graphs via node- and edge-level prediction. GDSS [Jo *et al.*, 2022] formulates graph generation as a continuous-time diffusion process based on stochastic differential equations, demonstrating the effectiveness of whole-graph diffusion for molecular generation. Diffusion models have also been explored in graph anomaly detection. [Ma *et al.*, 2024] generates node features via probabilistic denoising diffusion while keeping the adjacency matrix fixed, aiming to simulate anomalous perturbations. AGDiff [Cai *et al.*, 2025] addresses graph-level anomaly detection by modeling the generation of abnormal graphs, and DiffGAD [Li *et al.*, 2025] further integrates diffusion into the anomaly scoring mechanism. Despite progress, diffusion-based GAD methods remain conservative in structural modeling: some diffuse only node features without modifying connectivity, while others rely on feature-similarity–based adjacency reconstruction, which is fragile under distribution shifts. Moreover, general graph diffusion models neither explicitly model anomalies nor examine the co-evolution of edge density and feature diffusion, leaving principled edge addition in GAD largely unexplored.

3 Methodology

3.1 Preliminary and Problem Formulation

Given an attributed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X})$, where $\mathcal{V} = \{v_1, \ldots, v_N\}$ denotes the node set, $\mathcal{E}$ the edge set, and $\mathbf{X} \in \mathbb{R}^{N \times d}$ the node feature matrix. Let $\mathbf{A} \in \mathbb{R}^{N \times N}$ denote the adjacency matrix such that $A_{ij} = 1$ if $(v_i, v_j) \in \mathcal{E}$ and $A_{ij} = 0$ otherwise. The goal of self-supervised graph anomaly detection is to learn a function $\mathcal{F}_\theta : \mathcal{G} \to \mathbb{R}^N$ that assigns each node an anomaly score quantifying its deviation from normal patterns.

3.2 Training-free Generation Approach: LangGen

Langevin Dynamics. Langevin dynamics [Song and Ermon, 2019] enables sampling from high-probability regions of a

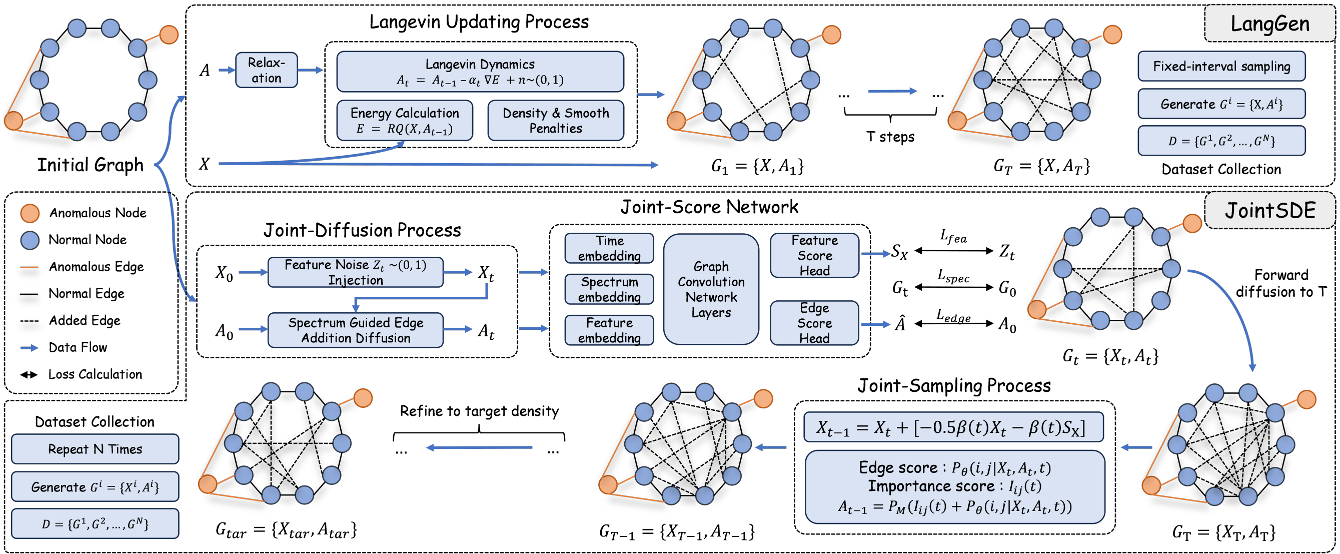

Figure 2: Illustration of the proposed LangGen and JointSDE frameworks. LangGen performs training-free, spectrum-guided structural completion by adding edges to the input graph based on an energy formulation, while keeping node features fixed. JointSDE jointly models the coupled evolution of node features and graph structure through a diffusion–reverse process, where spectral graph principles guide the synchronized generation of both the adjacency matrix and node representations.

target distribution $p_\theta(\mathbf{x})$ by iteratively combining gradient-driven updates with Gaussian noise:

$$\tilde{\mathbf{x}}_{t+1} = \tilde{\mathbf{x}}_t + \epsilon \nabla_{\mathbf{x}} \log p_\theta(\mathbf{x}) + \sqrt{2\epsilon}\,\mathbf{z}_t, \quad \mathbf{z}_t \sim \mathcal{N}(0, \mathbf{I}), \tag{1}$$

where ϵ denotes the step size. The gradient term drives samples toward regions of high probability, while the injected noise promotes exploration and prevents premature convergence to poor local minima. In the continuous-time limit, this process asymptotically samples from the target distribution.

Boltzmann Distribution and Energy-Based Formulation. The Boltzmann distribution establishes a principled connection between energy functions and probability densities:

$$p_\theta(\mathbf{x}) \propto \exp\big(-E_\theta(\mathbf{x})\big), \tag{2}$$

where $E_\theta(\mathbf{x})$ is a differentiable energy function. Under this formulation, the gradient of the log-density simplifies to

$$\nabla_{\mathbf{x}} \log p_\theta(\mathbf{x}) = -\nabla_{\mathbf{x}} E_\theta(\mathbf{x}), \tag{3}$$

enabling efficient sampling by directly optimizing the energy landscape without explicitly evaluating or normalizing the probability density $p_\theta(\mathbf{x})$.

Spectral Motivation for Graph Anomaly Detection. Recent studies [Tang *et al.*, 2022] have shown that anomalies exhibit elevated high-frequency energy in the graph spectral domain, and such accumulated energy can be captured by the Rayleigh Quotient (RQ):

$$RQ(\mathbf{X}, \mathbf{A}) = \frac{\mathrm{tr}(\mathbf{X}^\top L \mathbf{X})}{\mathrm{tr}(\mathbf{X}^\top \mathbf{X})} = \frac{\sum_{i,j} A_{ij} \|\mathbf{x}_i - \mathbf{x}_j\|^2}{\sum_i \|\mathbf{x}_i\|^2}, \tag{4}$$

where, $L = D - A$ denotes the combinatorial graph Laplacian, D is the degree matrix. Minimizing RQ promotes spectrally smooth structures while suppressing anomalous high-frequency patterns. By incorporating RQ into energy-based Langevin dynamics, the sampling process is explicitly guided toward spectrally coherent graph configurations. Notably, when the node feature matrix $\mathbf{X}$ is fixed during generation, energy optimization with respect to the adjacency matrix A directly governs structural evolution, marking a shift from conventional feature-generation paradigms to explicit edge generation. Under this setting, minimizing RQ favors the addition of structurally consistent edges among normal nodes rather than indiscriminate edge removal.

Energy Function Design. To enable gradient-based optimization over graph structure, we treat the adjacency matrix A as a diffusion variable and apply a continuous relaxation. Given an initial binary adjacency matrix $\mathbf{A}_0$, we first clamp entries to $[\epsilon, 1-\epsilon]$ to avoid logit singularities ($\epsilon = 10^{-6}$), then compute

$$\tilde{\mathbf{A}}_0 = \mathrm{sigm}\left(\frac{1}{\tau_t} \log \frac{\mathbf{A}_0}{1 - \mathbf{A}_0}\right), \tag{5}$$

where $\mathrm{sigm}(\cdot)$ is the sigmoid function and τ_t controls the relaxation temperature, allowing differentiable updates without introducing learnable parameters.

The total energy function is defined as

$$E_\theta(\mathbf{X}, \mathbf{A}) = RQ(\mathbf{X}, \mathbf{A}) + \lambda_{dens} \left\|\mathrm{Dens}(\mathbf{A}) - \rho_{\mathrm{tar}}\right\|^2 + \lambda_{smo} \|\mathbf{A} - \mathbf{A}_0\|^2, \tag{6}$$

where $\mathrm{Dens}(\mathbf{A}) = 2|\mathcal{E}|/N^2$ denotes graph density, ρ_{tar} is the target density, and $\lambda_\cdot$ are trade-off coefficients. The density and smoothness terms prevent degenerate solutions and stabilize structural evolution.

Langevin Update and Annealing. The Langevin update for adjacency matrix generation is given by

$$\tilde{\mathbf{A}}_{t+1} = \tilde{\mathbf{A}}_t - \epsilon \nabla_A E_\theta(\mathbf{X}, \mathbf{A}_t) + \epsilon \sigma_t \mathbf{z}_t, \quad \mathbf{z}_t \sim \mathcal{N}(0, \mathbf{I}), \tag{7}$$

where ϵ and σ_t follow annealing schedules to allow coarse exploration in early iterations and fine-grained refinement later. This process produces spectrally coherent, edge-augmented graphs without requiring any model training.

3.3 Feature-Structure Joint Diffustion: JointSDE

Joint Noising Process. The joint noising process characterizes the coupled degradation of node features and graph structure under a unified diffusion time scale, allowing structural uncertainty to adapt to the evolving feature geometry. As the diffusion time t increases, node features progressively lose discriminative information, while structural constraints are gradually relaxed, driving the system toward a highly stochastic state and providing diverse initial conditions for subsequent reverse refinement.

For continuous node features, we adopt a variance-preserving diffusion formulation [Song *et al.*, 2021b] with the closed-form forward process

$$\mathbf{X}_t = \alpha(t)\mathbf{X}_0 + \sigma(t)\mathbf{Z}, \qquad \mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \tag{8}$$

where $\alpha(t) \in (0, 1]$ controls signal preservation and $\sigma(t) = \sqrt{1 - \alpha^2(t)}$ denotes the noise scale. As t increases, $\mathbf{X}_t$ converges to an isotropic Gaussian.

For graph structure, we adopt a diffusion-inspired probabilistic relaxation that maps the discrete adjacency into time-dependent edge probabilities $P_{ij}(t) \in (0, 1)$:

$$P_{ij}(t) = \mathrm{sigm}\left(\frac{1}{\tau(t)} \log \frac{P_{ij}^{(0)}}{1 - P_{ij}^{(0)}} + S_{ij}(t)\right), \tag{9}$$

where $P_{ij}^{(0)}$ is obtained from the clamped $A_{0,ij}$ via logit, $\tau(t)$ controls temperature, and $\mathrm{sigm}(\cdot)$ is the sigmoid function. The final adjacency used for backbone training is binarized by thresholding at 0.5, with optional symmetry enforcement and self-loop handling.

The guidance term $S_{ij}(t)$ jointly encodes feature coherence and topological accessibility:

$$S_{ij}(t) = \frac{1}{2}\,\mathrm{Sim}_{ij}(t) + \frac{1}{2}\,\frac{R_{\mathrm{eff}}(i, j; \mathbf{A}_0)}{\max_{p,q} R_{\mathrm{eff}}(p, q; \mathbf{A}_0)}, \tag{10}$$

where $\mathrm{Sim}_{ij}(t) = \frac{1}{2}\big(\cos(\mathbf{x}_i(t), \mathbf{x}_j(t)) + 1\big)$ measures feature similarity under diffusion, and $R_{\mathrm{eff}}(i, j; \mathbf{A}_0) = L^+_{ii} + L^+_{jj} - 2L^+_{ij}$ denotes the effective resistance distance derived from the pseudoinverse of the original Laplacian, capturing global multi-path connectivity. This formulation enables a tightly coupled noising process in which structural uncertainty is continuously modulated by the evolving feature state.

Joint Score Network. The joint score network models feature scores and structural uncertainty on the noisy graph $\mathcal{G}_t = (\mathbf{X}_t, \mathbf{A}_t)$ under a shared diffusion time t, enabling coordinated reverse-time refinement of node attributes and graph topology. A unified encoder f_θ processes $(\mathbf{X}_t, \mathbf{A}_t, t)$ with explicit time conditioning, where the time embedding is defined as

$$\phi(t) = W_{t,2}\,\mathrm{SiLU}(W_{t,1}t + b_{t,1}) + b_{t,2}. \tag{11}$$

To capture global structural perturbations induced by structural noise, we construct the symmetric normalized Laplacian

$$L_t = I - D_t^{-1/2}\mathbf{A}_t D_t^{-1/2}, \tag{12}$$

and perform spectral decomposition $L_t = \Phi_t \Lambda_t \Phi_t^\top$. The first k low-frequency eigenvectors $\Phi_{t,k}$ provide spectral coordinates for each node. The initial node representation is given by

$$\mathbf{h}_i^{(0)} = \Big[\mathbf{X}_{t,i} W_e + b_e \parallel E_{\mathrm{spec}}(\Phi_{t,k}[i, :])\Big], \tag{13}$$

and updated through $\mathcal{L}$ layers of time-conditioned message passing:

$$\mathbf{h}^{(\ell)} = \mathrm{SiLU}\Big(\mathrm{LN}\Big(\tilde{D}_t^{-1/2}\tilde{A}_t\tilde{D}_t^{-1/2}\mathbf{h}^{(\ell-1)}W^{(\ell)} + \phi(t)\Big)\Big), \tag{14}$$

where $\tilde{A}_t = \mathbf{A}_t + I$ and $\tilde{D}_t$ is the corresponding degree matrix. The final node representations are denoted by $\mathbf{H} = \mathbf{h}^{(\mathcal{L})}$.

The feature score head predicts the denoising score $\mathbf{S}_X = g_{\mathrm{fea}}(\mathbf{H})$ and is trained via denoising score matching:

$$\mathcal{L}_{\mathrm{fea}} = \mathbb{E}_{\mathbf{X}_0, \mathbf{Z}, t}\big[\|\mathbf{S}_X + \tfrac{\mathbf{Z}}{\sigma(t)}\|^2\big], \tag{15}$$

where $\mathbf{Z}$ denotes standard Gaussian noise.

For structural denoising, we estimate edge existence likelihoods conditioned on the noisy graph as a tractable surrogate for reverse-time refinement. An edge-level representation is constructed as $\mathbf{u}_{ij} = \big[\mathbf{h}_i^{(\mathcal{L})} \parallel \mathbf{h}_j^{(\mathcal{L})} \parallel \phi(t)\big]$, and the edge head outputs $\hat{A}_{ij} = g_{\mathrm{edge}}(\mathbf{u}_{ij})$, with the objective

$$\mathcal{L}_{\mathrm{edge}} = -\sum_{i,j}\big(A_{0,ij}\log\hat{A}_{ij} + (1 - A_{0,ij})\log(1 - \hat{A}_{ij})\big). \tag{16}$$

To preserve global geometry during reverse refinement, we introduce a low-frequency spectral regularization

$$\mathcal{L}_{\mathrm{spec}} = \sum_{k=1}^{K}(\lambda_k^{\mathrm{gen}} - \lambda_k^{\mathrm{orig}})^2, \tag{17}$$

where λ_k^{gen} and $\lambda_k^{\mathrm{orig}}$ denote the k-th eigenvalues of the symmetric normalized Laplacian of the generated and original graphs, respectively. The joint score network is trained by minimizing

$$\mathcal{L}_{\mathrm{joint}} = \mathcal{L}_{\mathrm{fea}} + \mathcal{L}_{\mathrm{edge}} + \mathcal{L}_{\mathrm{spec}}, \tag{18}$$

enforcing consistency between feature-level score estimation and structure-level refinement under a shared latent representation.

Joint Sampling Process. Unlike conventional diffusion models that initialize sampling from pure Gaussian noise, JointSDE injects noise at a randomly sampled diffusion time t and performs partial reverse-time evolution. During sampling, node features are refined via a continuous-time reverse diffusion process, while graph structure evolves through a deterministic, feature-conditioned edge selection mechanism.

The reverse-time update of node features follows the VP–SDE and is discretized using an Euler scheme:

$$\mathbf{X}_{t-\Delta t} = \mathbf{X}_t + \big(-\tfrac{1}{2}\beta(t)\mathbf{X}_t - \beta(t)\mathbf{S}_X(\mathbf{X}_t, \mathbf{A}_t, t)\big)\,\Delta t, \tag{19}$$

where $\mathbf{S}_X$ denotes the learned feature score function.

Structural refinement is performed deterministically at each step by retaining or pruning edges according to a feature-conditioned priority score, avoiding the need to explicitly define a stochastic reverse process over discrete adjacency variables. For each edge (i, j), the structural importance score is defined as

$$I_{ij}(t) = \lambda_{\text{spec}}\, \tilde{S}_{ij}^{(\text{spec})} + \lambda_{\text{sim}}\, \tilde{S}_{ij}^{(x)} + \lambda_{\text{deg}}\, \tilde{D}_{ij}, \qquad (20)$$

where $\tilde{S}_{ij}^{(\text{spec})} = \sum_{k=1}^{K'} \omega_k \big(\phi_k(i) - \phi_k(j)\big)^2$, $\tilde{S}_{ij}^{(x)} = \text{sigm}\Big(\frac{\mathbf{x}_i^\top \mathbf{x}_j}{\|\mathbf{x}_i\|\,\|\mathbf{x}_j\|}\Big)$. Here, $\tilde{S}_{ij}^{(\text{spec})}$ captures the contribution of an edge to low-frequency Laplacian components, with frequency-decaying weights ω_k emphasizing global geometric stability, while $\tilde{S}_{ij}^{(x)}$ measures local feature homophily. The degree-based regularizer $\tilde{D}_{ij} = \frac{1}{d_i+d_j+1}$ suppresses hub-dominated structural bias.

Learned structural preferences are incorporated via the model-predicted edge retention probability

$$P_\theta(i, j \mid \mathbf{X}_t, \mathbf{A}_t, t) = \text{sigm}(e_\theta(i, j \mid \mathbf{X}_t, \mathbf{A}_t, t)), \qquad (21)$$

and the final edge priority is defined as

$$J_{ij}(t) = I_{ij}(t) + \lambda_{\text{score}}\, P_\theta(i, j \mid \mathbf{X}_t, \mathbf{A}_t, t). \qquad (22)$$

At each reverse step, the top M_t edges with the highest priority scores are retained:

$$\mathbf{A}_{t-\Delta t} = \mathcal{P}_{M_t}(J(t)), \qquad (23)$$

where $\mathcal{P}_{M_t}(\cdot)$ selects the M_t highest-scoring edges and M_t decreases monotonically as $t \to 0$. Overall, the joint sampling process is summarized as

$$(\mathbf{X}_{t-\Delta t}, \mathbf{A}_{t-\Delta t}) = \Psi\big((\mathbf{X}_t, \mathbf{A}_t), t\big), \qquad (24)$$

where Ψ denotes the coupled reverse-time operator combining stochastic feature refinement with spectrally guided deterministic structural evolution.

3.4 GAD Training with Synthetic Data

To enhance a given anomaly detector $\mathcal{F}_\theta$, we augment the original graph $\mathcal{G}$ with a set of generated graphs $\mathcal{D} = \{\mathcal{G}^1, \ldots, \mathcal{G}^{|\mathcal{D}|}\}$. For LangGen, $\mathcal{G}^i = \{X, A^i\}$ uses original node features with generated edges, whereas for JointSDE, $\mathcal{G}^i = \{X^i, A^i\}$ generates both node features and edges. The detector is further optimized on both the original and generated graphs by minimizing

$$\mathcal{L}_{\text{GAD}} = \mathcal{L}(\mathcal{F}_\theta(\mathcal{G}), \mathcal{G}) + \frac{1}{|\mathcal{D}|} \sum_{i=1}^{|\mathcal{D}|} \mathcal{L}\big(\mathcal{F}_\theta(\mathcal{G}^i), \mathcal{G}^i\big).$$

4 Experiments

In this section, we conduct extensive experiments to answer the following research questions: **Q1:** Can the proposed methods consistently improve state-of-the-art GAD models? **Q2:** How do individual components in the energy formulation and diffusion framework contribute to structure generation during sampling? **Q3:** How does the proposed approach compare with existing edge-based graph augmentation strategies? **Q4:** What is the computational efficiency of the proposed graph generation process? **Q5:** Does the proposed method preferentially promote edge formation among normal nodes?

Table 1: Statistics of datasets used in our experiments, where # denotes quantity, Dim. the feature dimension, #Ano. the number of anomalies, Avg. Deg. the average node degree, and Ratio the proportion of anomalous nodes.

Dataset	#Node	#Edge	Dim.	Avg.Deg.	#Ano.	Ratio (%)
Cora	2,708	11,060	1,433	4.1	70	2.5
Amazon	13,752	515,042	767	37.2	350	2.5
BlogCatalog	5,196	171,743	8,189	66.11	300	5.77
ACM	16,484	71,980	8,337	8.73	597	3.62
Citeseer	3,327	4,732	3,703	2.77	150	4.50
PubMed	19,717	44,338	500	4.50	600	3.04
Weibo	8,405	407,963	400	48.5	868	10.3
Reddit	10,984	168,016	64	15.3	366	3.3
Books	1,418	3,695	21	2.6	28	2.0
Enron	13,533	176,987	18	13.1	5	0.04
Disney	124	335	28	2.7	6	4.8
Facebook	1,081	55,104	576	50.97	25	2.31

4.1 Experimental Setup

Datasets. We evaluate our method on twelve widely used graph anomaly detection benchmarks, including six datasets with *organic anomalies* (Weibo, Reddit, Books, Enron, Disney, Facebook) and six with *injected anomalies* (Cora, Amazon, BlogCatalog, ACM, Citeseer, PubMed), with dataset statistics reported in Table 1 and originally collected from [Liu *et al.*, 2024b; Liu *et al.*, 2024c].

Baselines and Metrics. We compare against seven representative unsupervised or self-supervised GAD methods: DOMINANT[Ding *et al.*, 2019], DONE[Bandyopadhyay *et al.*, 2020], AnomalyDAE[Fan *et al.*, 2020], CoLA[Liu *et al.*, 2021], GADNR[Roy *et al.*, 2024], BOURNE[Liu *et al.*, 2024a], and CGADM[Wei *et al.*, 2025]. Performance is evaluated using AUROC and AUPRC.

Implementation Details. LangGen performs training-free edge generation via energy-guided Langevin sampling with 1000 iterations. JointSDE adopts a diffusion-based framework with 1000 forward noising steps and 500 reverse denoising steps. For each graph, ten augmented graphs are generated. The number of edges in each generated graph is fixed to 150% of the original. Baselines are implemented using PyGOD[Liu *et al.*, 2024b] or official code with default settings. Results are averaged over five runs. All experiments are conducted on a single NVIDIA RTX 4090 GPU.

4.2 Improvement on Anomaly Detection Performance (Q1)

Tables 3 and 2 summarize the performance changes of different detectors after incorporating additional generated graphs during training. Across six organic and six injected anomaly datasets, all evaluated baselines consistently achieve improvements in both AUROC and AUPRC, demonstrating that the proposed edge-augmentation-based strategy effectively enhances the discriminative capability of existing graph anomaly detection models. These gains are observed across diverse graph types, including citation, social, review, transaction, and email networks, indicating that the method generalizes well despite substantial differences in degree distributions, community structures, and edge semantics.

Across most experimental settings, the absolute improvements in AUROC are larger than those in AUPRC. This trend

Table 2: Node anomaly detection performance on six real-world benchmark datasets with organic anomalies. Results are reported as mean ± standard deviation (in %). Green and red values indicate improvements and drops over the corresponding backbone model, respectively.

Methods	Weibo		Reddit		Books		Enron		Disney		Facebook	
	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC
DOMINANT	55.37±18.00	11.55±5.68	56.12±0.06	3.73±0.02	39.39±2.66	1.56±0.03	51.92±0.66	0.08±0.00	51.55±3.04	5.56±0.57	11.04±2.03	1.31±0.02
+ *LanGen*	89.58±0.10	23.37±0.35	57.24±0.11	3.74±0.02	41.50±1.69	1.61±0.02	53.72±0.73	0.08±0.00	53.01±1.24	5.82±0.18	14.73±0.09	1.36±0.00
+ *JointSDE*	89.51±0.26	23.41±0.58	57.20±0.04	3.74±0.00	42.79±1.27	1.65±0.04	55.43±0.11	0.08±0.00	66.61±1.76	8.78±0.84	14.22±0.18	1.36±0.00
DONE	57.96±0.97	5.63±0.23	51.40±2.26	4.09±0.12	37.91±1.64	1.51±0.05	41.10±3.15	0.14±0.03	42.80±4.23	4.80±0.04	17.14±1.94	1.39±0.04
+ *LanGen*	84.59±0.57	30.40±0.24	57.51±1.85	4.14±0.07	40.97±3.15	1.73±0.16	42.72±0.93	0.16±0.08	43.22±4.50	4.89±0.17	23.65±2.33	3.38±2.65
+ *JointSDE*	84.61±0.16	30.45±0.21	57.18±0.55	4.09±0.08	42.01±2.49	1.82±0.25	41.68±0.60	0.11±0.03	43.56±0.26	4.88±0.03	23.45±0.66	2.11±0.46
AnomalyDAE	76.55±1.31	14.80±0.43	50.06±2.03	3.22±0.22	48.87±4.82	2.45±1.32	45.85±13.16	0.09±0.02	66.67±4.17	11.02±0.67	46.31±2.03	2.48±0.02
+ *LanGen*	78.99±0.95	15.36±0.85	55.69±3.47	3.91±0.41	61.86±2.01	3.08±0.18	70.50±2.22	0.09±0.00	72.36±0.24	12.55±0.23	51.69±0.52	2.57±0.06
+ *JointSDE*	85.30±0.92	19.52±1.47	56.41±2.48	4.03±0.26	59.22±6.50	2.88±0.87	69.38±2.62	0.09±0.01	71.16±2.22	11.95±0.39	53.00±0.50	2.63±0.02
CoLA	27.54±1.22	3.65±0.57	50.01±1.10	3.82±0.28	49.69±4.20	2.02±0.18	50.30±2.33	0.05±0.02	42.45±4.25	5.64±0.38	71.98±1.82	7.34±1.03
+ *LanGen*	30.60±1.00	5.78±1.18	52.01±0.64	3.99±0.27	52.95±4.43	4.17±1.27	62.19±10.27	0.12±0.04	54.46±3.72	6.72±0.55	78.66±1.66	6.24±0.56
+ *JointSDE*	30.20±2.25	4.28±0.46	54.04±2.65	4.07±0.33	57.68±0.98	2.48±0.11	63.09±2.54	0.06±0.01	59.49±2.74	10.07±2.36	81.36±0.71	8.35±1.68
GADNR	38.36±4.19	3.15±0.18	47.24±7.60	3.31±0.58	44.15±0.61	1.75±0.02	45.18±3.66	0.07±0.00	27.97±0.50	3.72±0.03	14.22±3.67	1.38±0.05
+ *LanGen*	47.42±1.24	3.82±0.18	55.96±0.36	3.74±0.01	50.59±1.96	1.96±0.04	53.46±0.46	0.08±0.00	50.19±2.82	7.47±1.26	27.40±17.53	1.65±0.45
+ *JointSDE*	62.55±1.70	5.54±0.43	55.94±0.32	3.75±0.01	45.61±0.45	1.75±0.03	72.24±1.56	0.11±0.00	49.77±6.61	5.49±0.66	47.77±8.28	2.17±0.42
BOURNE	89.00±1.08	8.37±1.02	56.54±0.81	3.71±0.08	65.49±4.08	2.49±0.00	55.59±5.40	0.05±0.01	73.81±5.84	8.20±2.01	86.77±1.20	3.57±0.26
+ *LanGen*	91.26±0.10	10.11±0.72	64.81±0.51	4.34±0.05	70.71±0.72	2.50±0.07	84.44±4.43	0.06±0.02	77.82±0.99	9.88±1.73	92.29±0.98	5.73±0.17
+ *JointSDE*	90.09±1.25	8.25±0.98	65.77±0.79	3.77±0.20	69.09±1.82	2.49±0.00	84.30±1.18	0.06±0.01	82.46±1.34	9.44±1.12	91.11±0.99	5.86±1.26
CGADM	92.54±3.02	33.13±2.04	65.68±1.01	7.28±0.64	51.40±10.82	8.59±3.28	59.17±8.66	0.08±0.10	46.49±7.34	6.00±1.35	96.06±0.28	62.19±0.36
+ *LanGen*	95.24±0.11	34.62±0.57	66.50±0.21	7.72±0.28	59.53±5.77	11.95±5.65	60.11±5.91	0.09±0.04	62.80±4.46	15.85±10.76	96.33±0.03	62.40±0.26
+ *JointSDE*	95.12±0.07	33.27±0.54	66.32±0.18	7.32±0.40	59.04±4.42	12.73±7.17	65.11±3.51	0.10±0.02	62.92±4.17	12.92±8.90	96.35±0.06	62.21±0.27

Table 3: Node anomaly detection performance on six real-world benchmark datasets with injected anomalies. Results are reported as mean ± standard deviation (in %). Green and red values indicate improvements and drops over the corresponding backbone model, respectively.

Methods	Cora		Amazon		BlogCatalog		ACM		Citeseer		PubMed	
	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC
DOMINANT	79.10±4.37	32.75±1.98	50.33±0.01	2.50±0.00	60.81±4.23	8.80±1.34	74.39±2.96	10.61±0.39	82.89±7.17	35.44±0.88	80.01±0.57	9.88±0.13
+ *LanGen*	87.36±0.15	36.35±0.10	50.35±0.01	2.51±0.00	68.71±2.03	11.27±0.82	79.99±0.00	11.38±0.01	91.12±2.15	36.36±0.12	81.03±0.66	10.08±0.11
+ *JointSDE*	87.84±0.03	36.33±0.13	50.42±0.01	2.51±0.00	68.68±0.71	11.22±0.32	80.02±0.72	11.35±0.00	91.42±0.02	36.40±0.04	80.80±0.11	9.96±0.10
DONE	70.10±2.40	23.71±0.42	60.84±1.29	5.36±0.26	44.94±0.64	5.01±0.08	64.59±0.34	7.42±0.30	64.83±0.84	20.69±1.41	87.70±0.22	20.97±0.28
+ *LanGen*	84.05±1.12	53.77±0.36	75.40±2.33	6.01±0.42	76.46±0.14	32.59±0.20	93.59±0.11	29.71±0.81	90.53±0.59	48.12±1.47	91.63±0.20	23.26±0.30
+ *JointSDE*	88.91±0.24	56.54±2.12	68.14±1.31	6.72±1.20	76.28±0.06	31.98±0.53	93.62±0.12	29.92±0.44	90.85±0.17	45.89±3.30	91.57±0.37	22.18±1.53
AnomalyDAE	51.05±12.71	13.20±5.94	77.39±0.01	5.48±0.06	48.87±13.78	18.89±9.56	52.47±12.00	13.69±7.09	60.57±12.46	17.83±7.66	73.87±0.27	23.83±0.88
+ *LanGen*	76.95±0.23	25.88±0.89	98.49±0.06	54.4±0.00	74.75±0.02	36.87±0.10	74.86±3.43	26.22±1.66	76.12±0.06	27.80±0.24	75.06±0.33	25.73±0.13
+ *JointSDE*	75.74±0.19	24.51±1.00	98.51±0.06	54.42±0.00	74.96±0.19	36.97±0.09	74.80±0.21	26.30±0.43	75.91±0.10	26.40±1.01	75.33±0.50	24.98±0.89
CoLA	55.39±1.53	6.38±0.42	61.00±1.09	9.60±0.35	53.28±1.19	8.30±0.28	50.16±0.85	4.62±0.38	54.05±2.06	5.41±0.51	70.08±0.24	13.99±0.43
+ *LanGen*	58.94±1.59	7.32±0.43	63.6±0.81	9.81±0.23	53.78±1.60	8.86±1.06	53.79±0.13	8.87±1.27	55.14±1.03	5.65±0.29	70.87±0.38	14.29±1.48
+ *JointSDE*	67.42±1.13	12.42±0.97	75.98±1.46	20.12±1.48	55.54±0.86	13.53±2.43	51.84±0.97	4.79±0.60	59.73±4.13	7.78±1.46	75.17±4.44	14.97±3.00
GADNR	72.57±1.32	26.96±2.52	49.57±0.46	2.49±0.03	53.98±0.76	6.81±0.09	71.56±0.20	10.88±0.44	74.07±0.63	32.54±1.24	73.95±0.18	9.40±0.09
+ *LanGen*	75.34±0.19	31.40±0.20	49.58±0.69	2.49±0.06	55.01±0.22	6.82±0.06	71.93±0.32	11.10±0.04	75.71±0.67	34.76±1.15	74.84±0.76	10.73±1.90
+ *JointSDE*	75.37±0.29	30.68±0.78	51.50±0.85	2.64±0.15	54.86±0.42	6.87±0.00	71.47±0.12	10.12±0.01	76.07±0.57	33.76±1.38	74.12±0.62	9.79±0.63
BOURNE	78.95±0.36	8.59±0.20	93.50±0.51	12.49±0.78	71.22±0.11	10.06±0.13	82.41±1.09	11.35±1.10	92.23±0.25	11.94±0.10	89.87±0.12	13.64±0.41
+ *LanGen*	80.76±0.98	8.94±0.20	94.23±0.55	13.82±0.83	72.35±0.35	10.93±0.59	83.11±0.39	11.79±0.92	94.20±0.18	14.15±0.37	90.77±0.35	13.98±1.06
+ *JointSDE*	79.11±0.36	7.94±0.42	94.42±0.16	12.73±0.39	72.66±0.23	10.29±0.44	82.94±0.39	12.44±1.01	93.87±0.61	13.73±0.48	91.08±0.19	13.99±0.46
CGADM	81.10±3.96	38.85±3.41	95.24±1.47	44.94±1.75	72.72±2.52	30.89±3.01	70.31±1.33	36.26±5.62	83.70±0.42	51.49±7.00	90.06±2.71	60.49±3.70
+ *LanGen*	84.18±0.39	42.04±1.77	96.31±0.00	45.78±0.98	75.30±0.00	33.02±0.00	71.26±0.81	41.44±0.72	84.45±0.57	58.76±0.81	91.50±1.02	62.46±1.57
+ *JointSDE*	84.28±0.02	42.87±0.97	96.31±0.40	46.49±0.41	75.29±2.14	33.01±0.68	71.55±0.78	43.20±0.92	84.77±0.15	57.63±3.36	91.24±3.31	62.33±0.17

is consistent with the strengthened modeling of normal nodes induced by edge augmentation: by promoting more coherent connections among normal nodes, the detector more accurately captures their feature and structural distributions, reducing false positives and improving the true negative rate, which is more directly reflected by the ROC metric. In contrast, AUPRC emphasizes anomaly-focused precision–recall behavior under severe class imbalance, leading to comparatively smaller gains.

Notably, even for datasets with relatively dense original structures, such as Weibo, Facebook, and BlogCatalog, the proposed strategy yields substantial improvements, with AUROC gains of up to 30% and AUPRC gains of up to 10% in certain settings. This suggests that the benefits extend beyond trivial edge completion and are largely attributable to the spectrum-energy-guided edge diffusion mechanism, which reinforces plausible intra-community connections while suppressing noisy inter-community links, thereby facilitating clearer structural decision boundaries.

4.3 Ablation Study (Q2)

We conduct ablation studies to examine whether the energy formulation $E_\theta(X, A)$ in Eq. 6 effectively guides edge generation in LangGen and whether JointSDE captures the joint evolution of node features and graph structure. For LangGen, we remove or replace individual energy components, including density regularization (*w/o dens*), smoothness regularization (*w/o smo*), both terms (*w/o d&s*), and the Rayleigh Quotient, where the RQ-based energy is replaced by a readout-based alternative (*w/o RQ*). For JointSDE, we ablate generated features (*w/o fea*) or structure (*w/o stru*), edge priority components in Eq. 22 (*w/o score*, *w/o spe*), and the diffusion-based sampling strategy by directly sampling from Gaussian noise (*w/o diff*).

We adopt CoLA as the backbone and report results in Ta-

Table 4: Ablation Result on LangGen and JointSDE.

Methods	Cora	Citeseer	Books	Facebook
CoLA	55.39±1.53	54.05±2.06	49.69±4.20	71.98±1.82
+LangGen	**58.94±1.59**	**55.14±1.03**	**52.95±4.43**	**78.66±1.66**
w/o den	57.11±2.34	52.41±2.20	50.66±3.90	76.63±3.26
w/o smo	58.83±1.57	52.37±2.41	49.0±7.67	74.50±3.21
w/o d&s	57.75±2.19	51.30±3.00	51.42±7.37	73.62±3.17
w/o RQ	56.70±2.06	49.08±7.67	46.57±4.73	74.68±3.51
+JointSDE	**67.42±1.13**	**59.73±4.13**	**57.68±0.98**	**81.36±0.71**
w/o fea	60.91±3.18	58.43±1.71	49.67±5.33	76.94±2.48
w/o stru	48.59±6.90	42.59±7.81	45.84±6.07	78.22±2.13
w/o score	64.09±3.09	48.65±9.39	52.99±6.06	77.29±3.29
w/o spe	63.73±5.16	45.98±10.04	47.89±7.22	77.68±3.19
w/o diff	47.21±4.31	48.85±3.07	50.72±7.23	78.03±2.31

Table 5: Comparison with Other Edge Augmentations.

Methods	Cora	Citeseer	Books	Facebook
CoLA	55.39±1.53	54.05±2.06	49.69±4.20	71.98±1.82
+ edge drop	52.50±0.71	50.48±1.44	50.85±1.05	69.22±0.88
+ edge add	51.63±0.72	49.61±0.72	50.80±1.12	72.32±0.62
+LangGen	58.94±1.59	55.14±1.03	52.95±4.43	78.66±1.66
+JointSDE	67.42±1.13	59.73±4.13	57.68±0.98	81.36±0.71

ble 4. For LangGen, removing any component consistently degrades performance, with *w/o RQ* yielding the weakest or even negative gains, underscoring the critical role of spectral energy in guiding meaningful edge addition, while density and smoothness terms further stabilize generation and improve robustness. For JointSDE, ablating either features or structure causes clear performance drops, confirming the necessity of joint feature–structure evolution. Although *w/o fea* underperforms the full model, it remains superior to CoLA due to the intermediate noising–denoising process. In contrast, *w/o diff* exhibits dataset-dependent behavior, and the comparison between *w/o spe* and *w/o score* shows that spectral guidance is essential for edge priority estimation, as learned scores alone fail to capture global structural evolution.

4.4 Edge Augmentation Comparison (Q3)

Table 5 compares our methods with random edge deletion and random edge addition, where both perturbation ratios are fixed at 10% following common practice; performance gains and degradations are highlighted in green and red, respectively.

Across datasets, both random edge deletion and random edge addition exhibit highly inconsistent behavior, improving performance on some datasets while degrading it on others, indicating that such heuristic perturbations fail to provide a stable inductive bias for graph anomaly detection and typically require dataset- and method-specific tuning.

In contrast, LangGen and JointSDE operate at a higher structural level and leverage spectral graph principles to guide edge evolution, yielding consistent and robust performance gains across datasets without extensive hyperparameter tuning, and thereby outperforming conventional edge-level augmentation strategies.

4.5 Computational Efficiency (Q4)

In LangGen, the dominant cost lies in computing the Rayleigh Quotient within the energy function, which can be optimized

Table 6: Sampling Time on All Datasets (in seconds).

Methods	Cora	Amazon	BlogCatalog	ACM	Citeseer	PubMed	Weibo	Reddit	Books	Enron	Disney	Facebook
LangGen	0.16	5.36	0.87	20.49	0.27	10.64	1.24	2.98	0.12	4.47	0.10	0.17
JointSDE	44.26	1929.28	160.50	372	67.70	390.62	438.22	961.51	16.54	1485.03	3.50	11.65

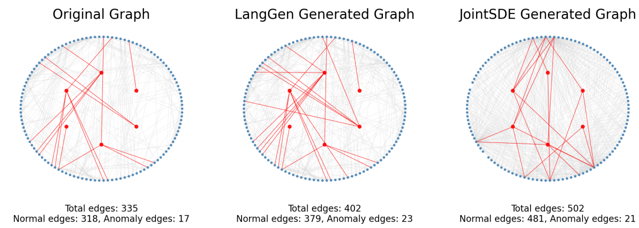

Figure 3: Visualization of graph structures before and after augmentation on the Disney dataset.

to $O(|\mathcal{E}|d)$ for a sparse graph with $|\mathcal{E}|$ edges and feature dimension d. In JointSDE, generation involves feature updates $O(Nd)$, edge importance scoring, and spectral decomposition, where computing the top k eigenvectors contributes $O(k|\mathcal{E}|)$ and score network inference over sampled edges $\mathcal{E}_{\text{sampled}}$ costs $O(|\mathcal{E}_{\text{sampled}}|d)$. Table 6 reports the average generation time per graph across all datasets.

4.6 Qualitative Analysis (Q5)

To examine whether our proposed method effectively increases connectivity among normal nodes, we visualize results on the Disney dataset, which was selected for its small size to facilitate clear and intuitive analysis. As shown in Figure 3, both LangGen and JointSDE generate substantially more edges compared to the original graph, whereas the number of edges involving anomalous nodes remains minimal. This visualization qualitatively confirms that our approach preferentially reinforces connections between normal nodes, highlighting its targeted enhancement capability.

4.7 More Experiments

Detailed descriptions of the method and experiments, comparisons with other augmentation approaches, qualitative analyses, validation on million-scale graphs, and sensitivity studies of hyperparameters and initialization appear in the appendix.

5 Conclusion and Future Work

This work demonstrates that moderate, principled edge addition can substantially improve graph anomaly detection, challenging the long-standing practice of relying on conservative edge dropping while deliberately avoiding edge addition. We introduce two self-supervised, spectrally guided augmentation approaches, in which graph structures are refined in a training-free manner via Langevin dynamics, or jointly evolved with node features through stochastic differential equations. Experiments on 12 benchmark datasets with 7 state-of-the-art GAD models show that the generated graphs consistently enrich connections among normal nodes while largely preserving anomalous structures. Future work will focus on reducing the number of hyperparameters and extending the framework to broader graph settings, including dynamic and structure-only graphs.

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A Table of Symbols

Table 7 summarizes the key symbols used in this paper along with their descriptions, providing readers with a convenient reference for understanding the overall processes of LangGen and JointSDE.

B Algorithms

Algorithms 1 and 2 summarize the detailed algorithmic procedures of LangGen and JointSDE, respectively. The pseudocode provides insight into their key differences, properties, and the detailed graph generation procedure.

B.1 LangGen Algorithm

Algorithm 1 LangGen: Training-free Spectral-Guided Edge Generation

Require: Attributed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X})$ with adjacency $\mathbf{A}_0$; step size ϵ; annealing schedules $\{\sigma_t\}_{t=1}^{T}$ and $\{\tau_t\}_{t=1}^{T}$; target density ρ_{tar}; trade-off coefficients $\lambda_{RQ}, \lambda_{dens}, \lambda_{smo}$

Ensure: Edge-augmented graph $\mathcal{G}' = (\mathcal{V}, \mathcal{E}', \mathbf{X})$

Adjacency Relaxation

Clamp $\mathbf{A}_0$ into $[\varepsilon, 1-\varepsilon]$ to avoid logit singularities

Compute relaxed adjacency

$$\tilde{\mathbf{A}}_0 = \text{sigm}\left(\frac{1}{\tau_0} \log \frac{\mathbf{A}_0}{1-\mathbf{A}_0}\right)$$

for $t = 0$ to $T-1$ **do**

  Energy Evaluation

  Compute Rayleigh quotient

$$RQ(\mathbf{X}, \mathbf{A}_t) = \frac{\text{tr}(\mathbf{X}^\top L_t \mathbf{X})}{\text{tr}(\mathbf{X}^\top \mathbf{X})}, \quad L_t = D_t - \mathbf{A}_t$$

  Compute total energy

$$E_\theta(\mathbf{X}, \mathbf{A}_t) = \lambda_{RQ} RQ + \lambda_{dens} \|\text{Dens}(\mathbf{A}_t) - \rho_{\text{tar}}\|^2 + \lambda_{smo} \|\mathbf{A}_t - \mathbf{A}_0\|^2$$

  Langevin Update

  Sample Gaussian noise $\mathbf{z}_t \sim \mathcal{N}(0, \mathbf{I})$

  Update relaxed adjacency

$$\tilde{\mathbf{A}}_{t+1} = \tilde{\mathbf{A}}_t - \epsilon \nabla_{\mathbf{A}} E_\theta(\mathbf{X}, \mathbf{A}_t) + \epsilon \sigma_t \mathbf{z}_t$$

end for

Discretization

Binarize $\tilde{\mathbf{A}}_T$ by thresholding at 0.5

Enforce symmetry and remove self-loops if required

return $\mathcal{G}' = (\mathcal{V}, \mathcal{E}', \mathbf{X})$

B.2 JointSDE Algorithm

C Implementation Details

This section provides a detailed description of the experimental settings, including the dataset types and characteristics, the core methodologies and implementation details of the baseline models, and the hyperparameter configurations of the proposed methods.

C.1 Dataset Details

The 12 graph anomaly detection datasets used in this study are collected from real-world sources and cover diverse graph types, including academic citations, e-commerce, and social networks. Based on the type of anomalies, the datasets can be divided into two categories: organic anomalies and injected anomalies.

Organic anomalies These datasets contain real anomalies occurring naturally in different applications such as online reviews, crowdsourcing platforms, and question-answering systems.

- **Weibo**: Collected from the Tencent-Weibo platform. Ground-truth anomalies are users involved in more than five suspicious events.
- **Reddit**: Comprises posts collected from a subreddit. Ground-truth anomalies are users banned from Reddit.

Table 7: Summary of Notations

Symbol	Description
$\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X})$	Attributed graph with node set $\mathcal{V}$, edge set $\mathcal{E}$, and node features $\mathbf{X}$
$\mathcal{V} = \{v_1, \ldots, v_N\}$	Set of N nodes
$\mathcal{E}$	Edge set of the graph
$\mathbf{X} \in \mathbb{R}^{N \times d}$	Node feature matrix with feature dimension d
$\mathbf{x}_i$	Feature vector of node v_i
$\mathbf{A} \in \mathbb{R}^{N \times N}$	Adjacency matrix of the graph
A_{ij}	Binary or relaxed indicator of edge existence between nodes i and j
$\mathbf{A}_0$	Original (observed) adjacency matrix
$\mathcal{F}_\theta : \mathcal{G} \to \mathbb{R}^N$	Graph anomaly detector producing node-wise anomaly scores
$p_\theta(\mathbf{x})$	Target probability density function parameterized by θ
$E_\theta(\mathbf{x})$	Energy function defining an energy-based model
ϵ	Langevin step size
$\mathbf{z}_t \sim \mathcal{N}(0, \mathbf{I})$	Gaussian noise injected at iteration t
$L = D - \mathbf{A}$	Combinatorial graph Laplacian
D	Degree matrix with $D_{ii} = \sum_j A_{ij}$
$RQ(\mathbf{X}, \mathbf{A})$	Rayleigh quotient measuring graph spectral smoothness
$\mathrm{tr}(\cdot)$	Matrix trace operator
ρ_{tar}	Target graph density
$\mathrm{Dens}(\mathbf{A})$	Graph density defined as $2\|\mathcal{E}\|/N^2$
$\lambda_{RQ}, \lambda_{dens}, \lambda_{smo}$	Trade-off coefficients in the energy function
$\tilde{\mathbf{A}}_t$	Continuously relaxed adjacency matrix at iteration t
τ_t	Temperature parameter controlling adjacency relaxation
σ_t	Noise scale in Langevin dynamics
$\alpha(t)$	Signal preservation coefficient in VP diffusion
$\sigma(t)$	Noise standard deviation in feature diffusion
$\mathbf{X}_t$	Noisy node features at diffusion time t
$P_{ij}(t)$	Time-dependent probability of edge (i, j)
$S_{ij}(t)$	Feature- and structure-guided edge modulation term
$\mathrm{Sim}_{ij}(t)$	Feature similarity between nodes i and j at time t
$R_{\mathrm{eff}}(i, j; \mathbf{A}_0)$	Effective resistance distance between nodes i and j
L^+	Moore–Penrose pseudoinverse of the Laplacian
f_θ	Joint score network with time conditioning
$\phi(t)$	Learnable time embedding function
L_t	Symmetric normalized Laplacian at diffusion time t
Φ_t, Λ_t	Eigenvectors and eigenvalues of L_t
$\mathbf{h}_i^{(\ell)}$	Node representation of node i at layer ℓ
$\mathcal{L}$	Number of message passing layers
$\mathbf{S}_X$	Predicted feature denoising score
$\mathcal{L}_{\mathrm{fea}}$	Feature score matching loss
$\mathcal{L}_{\mathrm{edge}}$	Edge reconstruction loss
$\mathcal{L}_{\mathrm{spec}}$	Spectral regularization loss
$\mathcal{L}_{\mathrm{joint}}$	Joint training objective of JointSDE
$\beta(t)$	Diffusion rate function in VP–SDE
$I_{ij}(t)$	Structural importance score of edge (i, j)
$J_{ij}(t)$	Final edge priority score combining heuristics and learned scores
$P_\theta(i, j \mid \mathbf{X}_t, \mathbf{A}_t, t)$	Model-predicted edge retention probability
$\mathcal{P}_{M_t}(\cdot)$	Operator selecting top-M_t edges
M_t	Number of retained edges at diffusion time t
$\Psi(\cdot)$	Coupled reverse-time feature–structure refinement operator
$\mathcal{D} = \{\mathcal{G}^i\}$	Set of generated synthetic graphs
$\mathcal{L}_{\mathrm{GAD}}$	Training loss of the anomaly detector with augmentation

Algorithm 2 JointSDE: Joint Feature–Structure Diffusion with Targeted Reverse Refinement

Require: Original graph $\mathcal{G}_0 = (\mathbf{X}_0, \mathbf{A}_0)$; target density ρ_{tar}; diffusion steps T; noise schedule $\{\beta_t\}_{t=1}^{T}$; score network $f_\theta = (s_\theta^{(x)}, s_\theta^{(a)})$
Ensure: Augmented graph $\mathcal{G}^{\mathrm{aug}} = (\mathbf{X}_{\mathrm{tar}}, \mathbf{A}_{\mathrm{tar}})$

Preprocessing:
Clamp $\mathbf{A}_0$ into $(\epsilon, 1-\epsilon)$ and apply logit transform
Compute original Laplacian $L_0 = D_0 - \mathbf{A}_0$ and its pseudoinverse L_0^{+}

Forward noising (training phase only):
Sample diffusion time $t \sim \mathcal{U}(1, T)$
Sample $\mathbf{Z}_x \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
$\mathbf{X}_t \leftarrow \alpha(t)\mathbf{X}_0 + \sigma(t)\mathbf{Z}_x$
for each edge (i, j) **do**
  Compute feature similarity $\mathrm{Sim}_{ij}(t)$
  Compute effective resistance $R_{\mathrm{eff}}(i, j; \mathbf{A}_0)$
  $S_{ij}(t) \leftarrow \frac{1}{2}\mathrm{Sim}_{ij}(t) + \frac{1}{2}\frac{R_{\mathrm{eff}}(i,j)}{\max R_{\mathrm{eff}}}$
  $P_{ij}(t) \leftarrow \sigma\Big(\frac{1}{\tau(t)}\mathrm{logit}(A_{0,ij}) + S_{ij}(t)\Big)$
end for
Obtain noisy adjacency $\mathbf{A}_t$ by Bernoulli sampling or relaxation

Joint score learning:
Encode $(\mathbf{X}_t, \mathbf{A}_t, t)$ using time-conditioned GNN
Predict feature score $\hat{\mathbf{S}}_X = s_\theta^{(x)}(\mathbf{X}_t, \mathbf{A}_t, t)$
Predict edge score $\hat{\mathbf{S}}_A = s_\theta^{(a)}(\mathbf{X}_t, \mathbf{A}_t, t)$
Optimize $\mathcal{L}_{\mathrm{fea}} + \mathcal{L}_{\mathrm{edge}} + \mathcal{L}_{\mathrm{spec}}$

Joint reverse sampling (generation phase):
Sample initial time $t_0 \sim \mathcal{U}(t_{\min}, T)$
Initialize $\mathbf{X}_{t_0}, \mathbf{A}_{t_0}$ using forward noising
Set target edge count $M_{\mathrm{tar}} = \rho_{\mathrm{tar}} \cdot |\mathcal{E}_0|$
Define monotonic edge schedule $\{M_t\}$ with $M_{t_0} \gg M_{\mathrm{tar}}$
for $t = t_0, t_0 - \Delta t, \ldots, t_{\mathrm{tar}}$ **do**
  Feature reverse update:
  $\hat{\mathbf{S}}_X \leftarrow s_\theta^{(x)}(\mathbf{X}_t, \mathbf{A}_t, t)$
  $\mathbf{X}_{t-\Delta t} \leftarrow \mathbf{X}_t + \Big(-\frac{1}{2}\beta(t)\mathbf{X}_t - \beta(t)\hat{\mathbf{S}}_X\Big)\Delta t$
  Edge priority computation:
  for each candidate edge (i, j) **do**
    Compute spectral term $\tilde{S}_{ij}^{(\mathrm{spec})} = \sum_{k=1}^{K'} \omega_k(\phi_k(i) - \phi_k(j))^2$
    Compute feature homophily $\tilde{S}_{ij}^{(x)} = \sigma\Big(\frac{\mathbf{x}_i^\top \mathbf{x}_j}{\|\mathbf{x}_i\|\|\mathbf{x}_j\|}\Big)$
    Compute degree regularization $\tilde{D}_{ij} = \frac{1}{d_i + d_j + 1}$
    Predict learned edge score $P_\theta(i, j) = \sigma(e_\theta(i, j))$
    Compute joint priority $J_{ij}(t) = \lambda_{\mathrm{spec}}\tilde{S}_{ij}^{(\mathrm{spec})} + \lambda_{\mathrm{sim}}\tilde{S}_{ij}^{(x)} + \lambda_{\mathrm{deg}}\tilde{D}_{ij} + \lambda_{\mathrm{score}}P_\theta(i, j)$
  end for
  Targeted structure refinement:
  Retain top-$M_{t-\Delta t}$ edges according to $J_{ij}(t)$
  $\mathbf{A}_{t-\Delta t} \leftarrow \mathcal{P}_{M_{t-\Delta t}}(J(t))$
end for

Post-processing:
Remove self-loops if required
return $\mathcal{G}^{\mathrm{aug}} = (\mathbf{X}_{\mathrm{tar}}, \mathbf{A}_{\mathrm{tar}})$

- **Books**: A co-purchase network of books on Amazon with attributes similar to the Disney dataset. Ground-truth labels are derived from the 'amazonfail' tag information.
- **Enron**: An email network where each email is a node, and edges represent messages between addresses. Nodes sending spam are labeled as anomalies. Each node has 20 attributes summarizing content length, recipient counts, and time intervals.
- **Disney**: A co-purchase network of movies with attributes including price, rating, and number of reviews. Ground-truth anomalies are manually labeled by high school students via majority vote.
- **Facebook**: A social network where nodes represent users and edges denote friendship connections, reflecting localized community structures.

Injected anomalies These anomalies are synthetically generated by copying or modifying node attributes within the same graph, following the PyGOD library.

- **Cora**: A scientific citation network with publications as nodes and citation links as edges.
- **Amazon**: A segment of the Amazon co-purchase graph, where nodes are goods and edges indicate frequently co-purchased items. Node features are BoW-encoded reviews, and class labels correspond to product categories.
- **BlogCatalog**: An online blog-sharing network. Edges indicate follower-followee relations, and node attributes are user tags.
- **ACM**: A scientific citation network with publications as nodes and citation relations as edges.
- **Citeseer**: A citation network covering domains such as computer science and biomedicine.
- **PubMed**: Another citation network covering computer science and biomedical publications.

C.2 Baseline Implementations

We evaluate the effectiveness of the generated graphs using seven state-of-the-art graph anomaly detection (GAD) models, including DOMINANT, DONE, AnomalyDAE, CoLA, GADNR, BOURNE, and CGADM. These methods cover a broad spectrum of paradigms, such as autoencoder-based reconstruction, encoder–decoder architectures, reconstruction-error assumptions, contrastive self-supervised learning, and diffusion-based approaches.

For DOMINANT, DONE, AnomalyDAE, CoLA, and GADNR, we directly adopt the implementations provided in the PyGOD library and retain their default initialization and hyperparameter settings to ensure a fair comparison. BOURNE and CGADM are implemented using their publicly available official codebases, likewise without modifying any hyperparameters.

Below we briefly summarize the core ideas of each baseline.

- **DOMINANT** is an early representative GAD method based on a graph convolutional autoencoder framework. It adopts an encoder–decoder architecture with graph convolutional networks (GCNs) to jointly reconstruct node attributes and graph structure. Node anomaly scores are computed as a weighted combination of attribute reconstruction error and structural reconstruction error.
- **DONE** employs separate multi-layer perceptron (MLP) autoencoders to reconstruct node attributes and adjacency information, respectively. Structural and attribute representations are learned jointly, and anomaly scores are optimized together with node embeddings under a unified reconstruction objective.
- **AnomalyDAE** utilizes dual autoencoders for structure and attributes. The model encodes structural and attribute information into two latent spaces and reconstructs node attributes by integrating both embeddings, enabling the detection of anomalies that deviate in either structure or attributes.
- **CoLA** is the first contrastive self-supervised learning framework for GAD. It constructs positive and negative node–subgraph pairs through stochastic sampling and leverages contrastive objectives to capture localized structural and contextual inconsistencies for anomaly detection.
- **GADNR** focuses on neighborhood reconstruction by modeling both structural relations and attribute consistency within local neighborhoods. Anomalies are identified based on deviations in reconstructed neighborhood patterns.
- **BOURNE** proposes a unified bootstrap self-supervised framework that detects anomalies at both node and edge levels. It jointly leverages graph and hypergraph representations to enhance robustness against noise and structural perturbations.
- **CGADM** incorporates a prior-guided diffusion process for anomaly detection. It integrates a conditional anomaly estimator into both forward and reverse diffusion steps, allowing anomaly-aware guidance during the denoising process to improve detection accuracy.

C.3 Hyperparameter Settings

Several important hyperparameters are introduced in the proposed methods. For LangGen, two method-specific hyperparameters are defined in Eq. (6), which control the density regularization and smoothness terms. We set $\lambda_{dens} = 10$ and $\lambda_{smo} = 0.1$, respectively. For JointSDE, the method-specific hyperparameters are defined in the edge importance scoring function in Eq. (20), with $\lambda_{spec} = 0.4$, $\lambda_{sim} = 0.4$, and $\lambda_{deg} = 0.2$. For hyperparameters shared by both methods, such as the target density of the generated graphs, we set $\rho_{tar} = 1.2$ for both methods. All hyperparameter values are determined based on systematic hyperparameter analysis, as discussed in the hyperparameter analysis section.

D Additional Experiments

Table 8: Hyperparameter tuning configurations used in sensitivity analysis for LangGen and JointSDE.

Hyperparameter	Tuning Range / Candidates	Description
λ_{dens}	$\{1.0, 5.0, 10.0, 15.0, 20.0\}$	Weight of the density regularization term in LangGen
λ_{smo}	$\{0.1, 0.2, 0.5, 0.7, 1.0\}$	Weight of the smoothness regularization term in LangGen
λ_{spec}	$\{0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8\}$	Spectral contribution in edge importance scoring (JointSDE)
λ_{sim}	$\{0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8\}$	Feature similarity contribution in edge importance scoring (JointSDE)
λ_{deg}	$\{0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8\}$	Degree-based regularization weight in JointSDE
ρ_{tar}	$\{1.1, 1.2, 1.3, 1.5\}$	Target density ratio of generated graphs
$\|\mathcal{D}\|$	$\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$	Number of generated graphs used for augmentation

This section provides a comprehensive analysis of the stability and effectiveness of the proposed methods under various settings. Following the experimental plan summarized in Table 8, we conduct a systematic study of the hyperparameter sensitivity of LangGen and JointSDE, analyze their performance under different target graph densities, and examine the effect of varying the number of generated graphs on overall detection performance.

In addition, as the main text considers random edge dropping and random edge addition only at a fixed perturbation ratio of 10%, we further investigate a broader range of edge augmentation strategies and perturbation ratios, and compare them against the proposed methods.

D.1 Hyperparameter Sensitivity Analysis

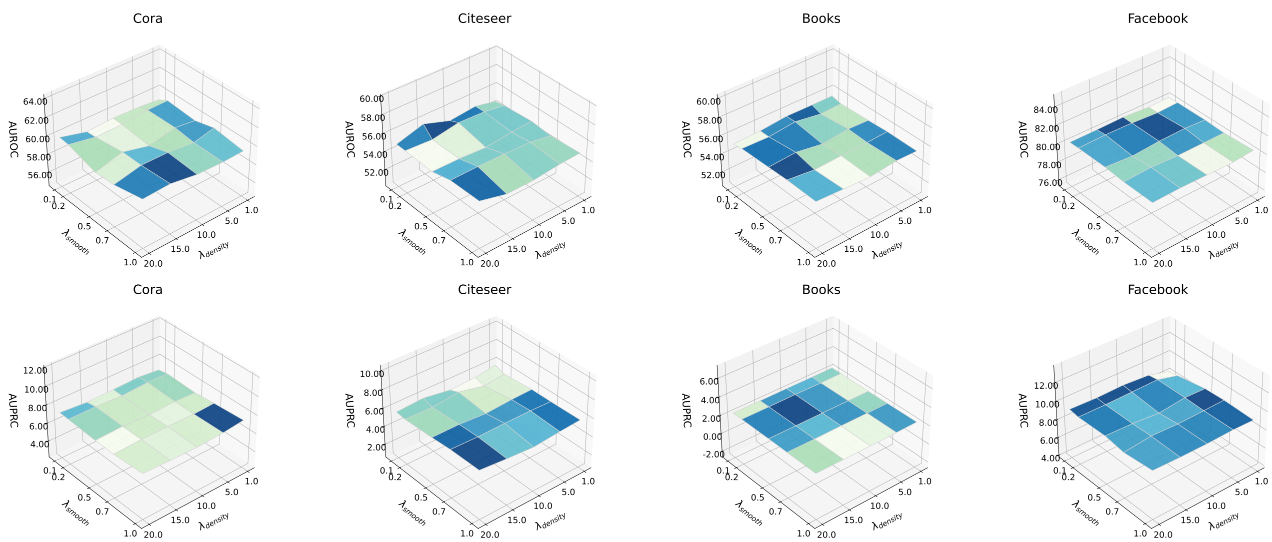

Figure 4: Anomaly detection performance in terms of AUROC and AUPRC on Cora, Citeseer, Books, and Facebook under different values of λ_{dens} and λ_{smo}. The x-axis and y-axis correspond to λ_{dens} and λ_{smo}, respectively, while the z-axis represents the corresponding performance metric.

Analysis on LangGen. To evaluate the impact of key hyperparameters on the quality of augmented graphs generated by LangGen, we conduct a sensitivity analysis on the energy function balance parameters λ_{dens} and λ_{smo}. Figure 4 presents CoLA AUROC across different parameter values. The results show that datasets exhibit different responses to the parameter settings, indicating that the optimal balance between density and smoothness depends on network type and dataset characteristics. Nevertheless, these variations do not lead to substantial performance changes; while some datasets show up to 1% fluctuation in AUROC, such deviations remain within a reasonable range for real-world datasets.

To avoid dataset- or method-specific hyperparameter tuning, we empirically fix $\lambda_{dens} = 10$ and $\lambda_{smo} = 0.1$ for LangGen, as this combination achieves satisfactory performance and efficiency across most datasets and settings. This configuration is adopted for practical convenience, while further improvements remain an avenue for readers to explore.

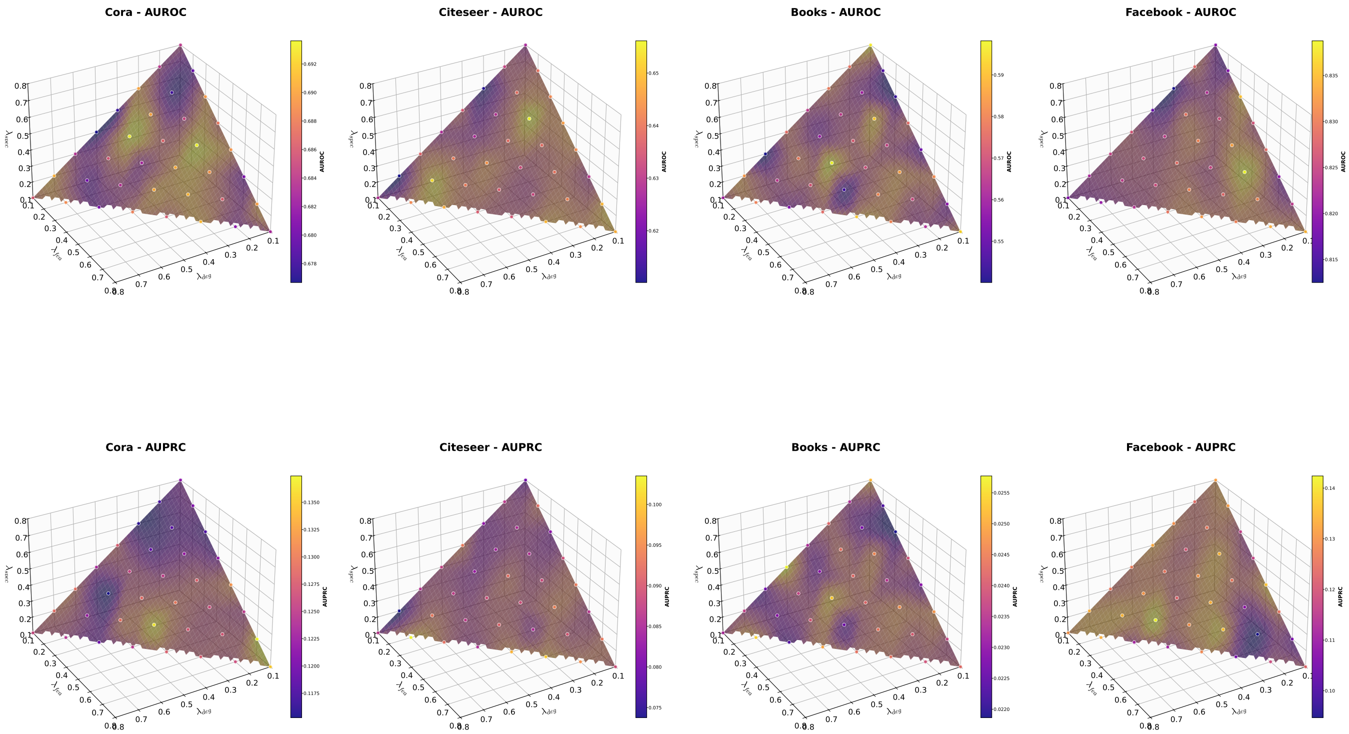

Figure 5: Sensitivity analysis of anomaly detection performance for JointSDE. The x-, y-, and z-axes correspond to λ_{spec}, λ_{sim}, and λ_{deg}, respectively, while color intensity on the 3D surface indicates the corresponding AUROC or AUPRC values, with brighter colors representing higher performance.

Analysis on JointSDE. We conduct a sensitivity analysis of the edge importance scoring function in JointSDE with respect to the spectral energy weight λ_{spec}, feature similarity weight λ_{sim}, and degree regularization weight λ_{deg}. In our experiments, we impose the constraint $\lambda_{spec} + \lambda_{sim} + \lambda_{deg} = 1$, which restricts the feasible parameter space to a simplex. Consequently, not all parameter combinations are explored, and the resulting performance landscapes correspond to a constrained subspace rather than an unconstrained three-dimensional grid.

Figure 5 visualizes the sensitivity results under this constraint. Variations in λ_{spec} and λ_{sim} implicitly determine λ_{deg}, and the color of each sampled point indicates the corresponding performance metric. The results show that extreme configurations—such as assigning a disproportionately large weight to one component while heavily suppressing the others—consistently fail to achieve optimal performance. This observation suggests that the spectral, similarity-based, and degree-based components jointly contribute to reliable edge importance estimation, and that their balanced integration is critical for effective graph generation.

In addition, we observe that different datasets favor different regions of the parameter space. This behavior aligns with the sensitivity patterns observed in the LangGen analysis and indicates that dataset characteristics influence the relative importance of spectral, feature-based, and structural cues in JointSDE.

For simplicity and robustness across datasets, we adopt $\lambda_{spec} = 0.4$, $\lambda_{sim} = 0.4$, and $\lambda_{deg} = 0.2$ as the default configuration, as this setting yields consistently competitive performance in most cases. It is worth noting, however, that this choice does not correspond to the optimal configuration for all datasets. Several alternative parameter combinations achieve superior performance, indicating that JointSDE admits further room for refinement. We leave a more exhaustive exploration of this parameter space to interested readers.

D.2 Effect of Target Density

Figures 6 and 7 illustrate the impact of varying target graph densities ρ_{tar} on anomaly detection performance for LangGen and JointSDE, respectively. As the target density increases, the AUROC scores on all four evaluated datasets exhibit a generally increasing trend. Moreover, compared with the original detector performance without augmentation reported in Tables 3 and 2, even the weakest results among the evaluated densities consistently outperform the unaugmented baselines. This observation supports our central finding that principled edge addition can effectively enhance graph anomaly detection performance.

Beyond the overall improvement trend, different target densities yield varying degrees of performance gains across datasets. This behavior suggests that both LangGen and JointSDE are able to selectively identify and reinforce connections among normal nodes within the same communities, rather than indiscriminately densifying the graph structure, thereby further validating the effectiveness of the proposed augmentation strategies.

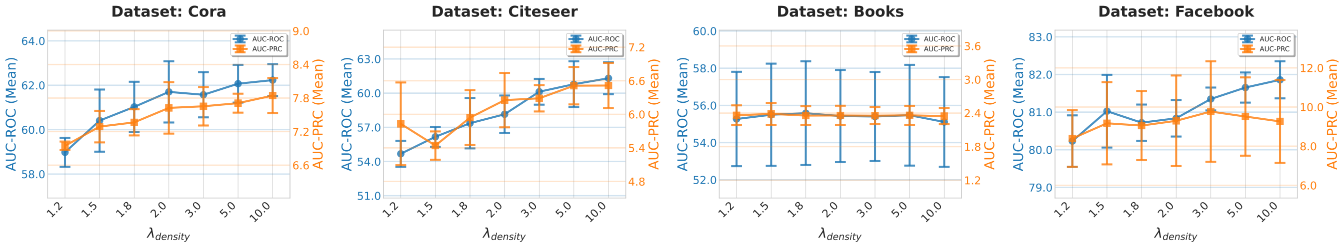

Figure 6: Anomaly detection performance under different target graph densities ρ_{tar} when using LangGen for graph augmentation. Reported results are mean AUROC or AUPRC values, with error bars indicating standard deviations over repeated runs.

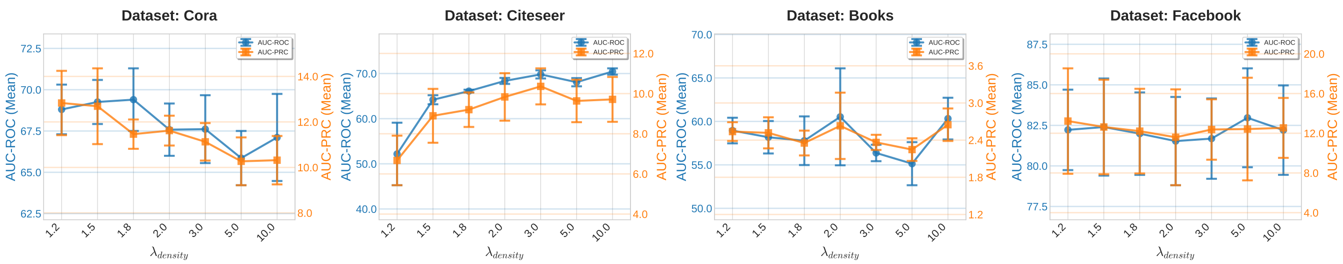

Figure 7: Anomaly detection performance under different target graph densities ρ_{tar} when using JointSDE for graph augmentation. Reported results are mean AUROC or AUPRC values, with error bars indicating standard deviations over repeated runs.

Based on these observations, we adopt $\rho_{tar} = 1.5$ as a default setting for practical evaluation, as it lies within a regime where performance improvements are consistently observed while maintaining a moderate computational overhead. Although this choice provides a favorable balance between efficiency and accuracy across most settings, other density configurations can lead to even stronger performance gains. This indicates that the proposed methods offer substantial room for further exploration through more fine-grained density selection when higher performance is desired.

D.3 Effect of the Number of Generated Graphs

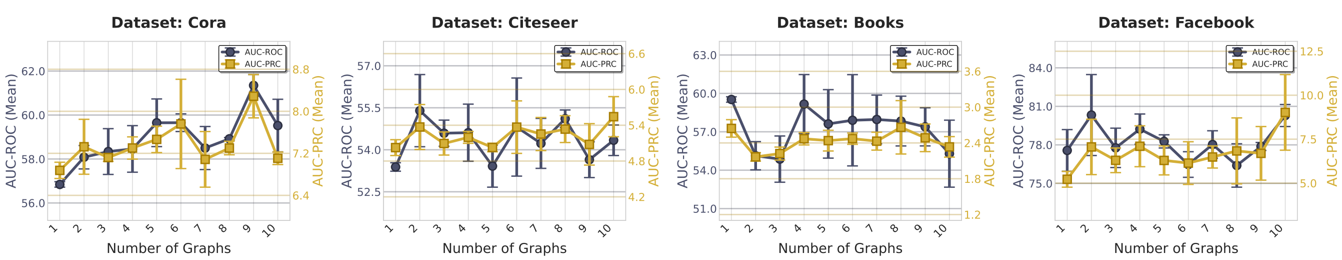

Figure 8: Impact of the number of graphs generated by LangGen on anomaly detection performance.

In the main experiments, we use 10 generated graphs per dataset from LangGen and JointSDE for augmentation, except for PubMed, where only 3 graphs are employed due to memory constraints. To investigate whether increasing the number of generated graphs can further improve detector performance, we conduct additional experiments on four datasets.

As shown in Figures 8 and 9, the effect of increasing the number of generated graphs on anomaly detection performance is dataset- and method-dependent. A small number of augmented graphs generally yields limited improvement, while adding more graphs does not consistently lead to further gains. These results indicate that the optimal number of augmented graphs varies across datasets and methods, reflecting differing preferences in augmentation for each setting.

D.4 Additional Edge Augmentation Comparisons

To further evaluate the effectiveness of our proposed edge addition augmentation methods, we compare them with other edge-related data augmentation strategies. Specifically, we consider the following baselines: random edge deletion (+edge drop), random edge addition (+edge add), feature similarity-based kNN (+KNN), and Personalized PageRank (PPR). For each method, we evaluate perturbation ratios ranging from 0.1 to 0.5.

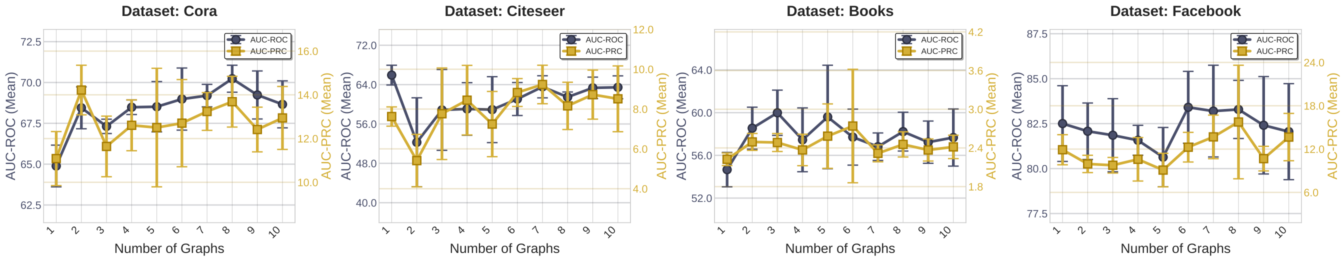

Figure 9: Impact of the number of graphs generated by JointSDE on anomaly detection performance.

Table 9: Comparison with other edge augmentations. Edge drop refers to random edge deletion, edge add refers to random edge addition, kNN denotes feature similarity-based k-nearest neighbors, and PPR stands for Personalized PageRank.

Methods	Ratio	Cora	Citeseer	Books	Facebook
CoLA	- -	55.39±1.53	54.05±2.06	49.69±4.20	71.98±1.82
+ edge drop	0.1	52.50±0.71	50.48±1.44	50.85±1.05	69.22±0.88
	0.2	51.94±1.30	51.77±1.42	52.38±3.44	72.47±2.71
	0.3	51.91±1.43	52.81±1.74	52.07±3.25	71.53±2.77
	0.4	52.48±1.48	52.16±3.06	52.53±4.27	70.79±3.27
	0.5	53.06±1.21	51.77±3.72	52.63±4.25	69.09±3.37
+ edge add	0.1	51.63±0.72	49.61±0.72	50.80±1.12	72.32±0.62
	0.2	47.50±4.08	49.49±2.17	51.79±3.80	72.07±1.96
	0.3	47.57±4.20	44.38±2.86	51.79±3.51	72.55±4.62
	0.4	41.41±2.45	43.71±2.97	51.87±3.70	67.59±4.29
	0.5	40.13±2.23	40.48±1.73	51.99±4.46	63.06±7.79
+ kNN	0.1	51.39±1.42	50.99±1.69	51.42±4.06	70.22±2.48
	0.2	53.28±1.56	52.27±1.79	51.33±3.99	67.39±2.72
	0.3	52.20±1.49	52.43±1.94	51.31±4.09	66.58±2.47
	0.4	52.51±1.81	52.19±1.99	51.28±4.29	65.42±2.17
	0.5	51.33±2.57	52.17±1.42	51.10±4.15	65.15±2.41
+ PPR	0.1	53.06±2.04	49.41±1.43	51.79±3.49	54.95±4.73
	0.2	53.12±1.81	50.89±1.53	51.64±3.64	52.08±3.76
	0.3	53.48±1.22	50.67±0.92	51.80±3.60	50.87±3.30
	0.4	54.12±1.98	51.83±1.44	51.55±3.76	49.54±3.47
	0.5	53.97±1.58	52.39±1.18	51.45±3.91	48.60±3.63
+LangGen	- -	58.94±1.59	55.14±1.03	52.95±4.43	78.66±1.66
+JointSDE	- -	67.42±1.13	59.73±4.13	57.68±0.98	81.36±0.71

The kNN method relies on the assumption that nodes similar in the feature space are more likely to belong to the same community or share the same semantic role. By adding edges between nodes based on feature similarity, this approach alleviates graph sparsity, reinforces homophily, and improves anomaly detection performance that depends on both structural and feature information. Mathematically, the kNN edges for node i are defined as

$$Nk(i) = \mathrm{TopK}_{j \neq i}\ \mathrm{sim}(i, j),$$

where $\mathrm{sim}(i, j)$ denotes the similarity between nodes i and j in the feature space.

In contrast, PPR focuses on higher-order structural similarity. Even if two nodes are not directly connected, they should be linked if they are "easily reachable" under a random walk perspective. Compared with kNN, PPR does not rely on node features, captures multi-hop structural relationships, and is particularly effective for low-degree nodes and sparse graphs. The PPR vector for node i is defined as

$$\pi_i = \alpha e_i + (1 - \alpha)\pi_i P, \quad P = D^{-1}A,$$

where D is the diagonal degree matrix and A is the adjacency matrix, and the formula is typically computed iteratively until convergence.

The experimental results are presented in Table 9. These baseline methods do not achieve consistent improvements across all datasets; in many cases, performance even degrades. This observation indicates that accurately capturing the relationships among normal nodes and within-community nodes requires careful and principled modeling of both feature and structural information. In contrast, our proposed LangGen and JointSDE methods effectively address these challenges, improving anomaly detection performance while avoiding dataset-specific hyperparameter tuning, which highlights the broad applicability and robustness of our approach.